

Proactive Memory for Event-Driven LLM Agents

Anonymous ACL submission

Abstract

Proactive LLM agents anticipate service opportunities from ongoing observations and accumulated history. A central challenge is determining which historical observations remain applicable to the current event and can support a timely service decision. We propose Proactive Memory (ProactMem), combining bounded recent events, source-checked daily trajectories, and replay-assessed temporal patterns. Daily validation checks extracted facts against source events, while historical replay assesses recurring patterns before they are matched to current events. We also construct ProactiveStream, a longitudinal benchmark covering three application scopes, where agents process isolated event streams chronologically under episode-aware opportunity labels. ProactMem improves Effectiveness by 4.3–15.7 percentage points over the strongest external baseline in each scope. Ablation studies support the contributions of hierarchical memory organization and validation to service-opportunity detection.

1 Introduction

Proactive LLM agents anticipate useful assistance as a user’s situation evolves, taking initiative before an explicit request. Recent systems study when to propose a task, detect emerging needs, and use contextual or historical information to guide that judgment (Lu et al., 2025; Yang et al., 2025; Xie et al., 2026; Li et al., 2026). Across our Sports, Home, and Code scopes, this requires interpreting a continuing stream of observations and deciding whether a service opportunity warrants attention now.

Historical context is central to this decision. Figure 1 illustrates how earlier evening walks can inform a service judgment when current activity is low and the user is available. Daily trajectories and temporal patterns retain this context beyond a

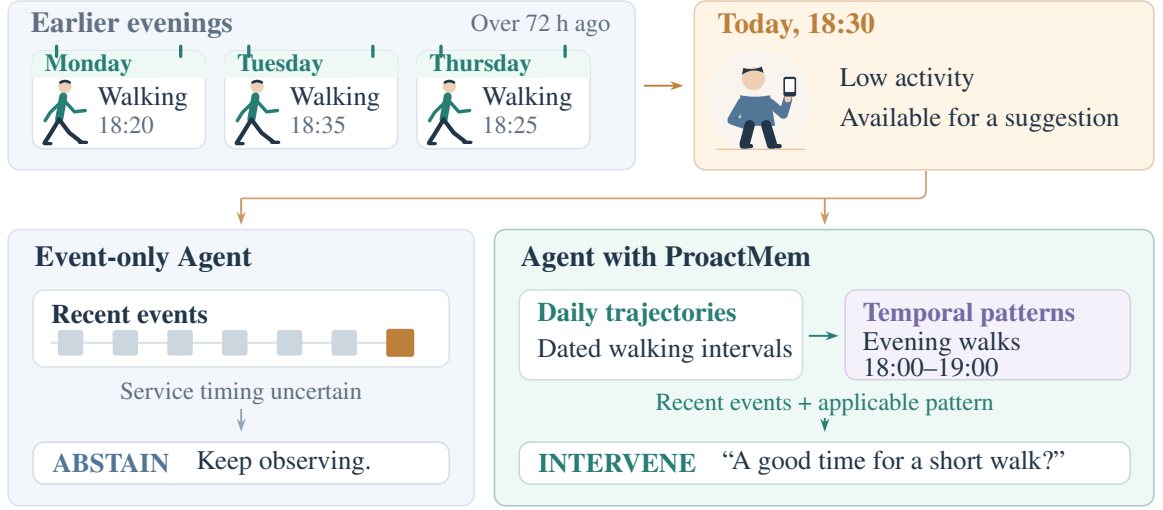
recent-event window. The agent must connect that history to a concrete, timely service.

Agent memory systems retain observations, consolidate experience, and retrieve related information (Park et al., 2023; Packer et al., 2023; Chhikara et al., 2025; Xu et al., 2025). To turn this history into decision context, proactive agents must assess whether an abstraction reflects its source events, whether a proposed temporal pattern has recurring support, and whether that pattern applies now. Our central question is *how to turn longitudinal history into reliable, currently applicable context for proactive service decisions*.

We propose Proactive Memory (ProactMem; Figure 2), a hierarchical memory framework that organizes event histories into source-checked daily trajectories and replay-assessed temporal patterns for proactive service decisions. Bounded Event Memory preserves recent observations, while Daily Trajectory Memory retains source-checked abstractions beyond the recent window. Temporal Pattern Memory consolidates recurring conditions from these daily trajectories and assesses their historical support through replay. At decision time, the agent combines recent events and daily trajectories with currently applicable patterns. The decision LLM assesses whether current observations or historical context support a concrete, timely, low-risk service; episode control tracks whether the proposed opportunity has already been covered.

We construct *ProactiveStream* to study these decisions over 4,539 events from 17 isolated streams. Its three scopes preserve causal history and define episode-aware reference opportunities, allowing evaluation of memory formation and repeated service decisions within the same streams. ProactMem improves Effectiveness by 4.3–15.7 percentage points over the strongest of five external baselines in each scope. A separate recursive-summary comparison and targeted ablations examine the roles of historical organization and support checks.

From past activity to timely assistance



Illustrative example

Figure 1: Illustrative Sports scenario. Both agents receive the same current event. ProactMem supplements recent events with daily trajectories and temporal patterns built from earlier walking observations, informing the assessment of a timely activity suggestion. The two responses depict possible decisions in this conceptual example.

Our contributions are:

- **Method.** ProactMem combines source-checked daily abstraction, replay-based pattern assessment, and selection of currently applicable patterns to supply historical context for proactive service decisions.
- **Benchmark.** ProactiveStream supports longitudinal evaluation of memory formation and repeated service decisions, preserving causal history, entity isolation, and episode-aware reference opportunities across three scopes.
- **Empirical findings.** ProactMem achieves higher Effectiveness than the evaluated baselines and summary control in all three scopes; targeted ablations support the contributions of hierarchical organization and validation.

2 Problem Formulation

Let u denote an independently evaluated entity, such as a participant, household, or repository, with a time-ordered public event stream

$$E^u = (e_1^u, \dots, e_{T_u}^u), \quad e_t^u = (\tau_t^u, x_t^u), \quad (1)$$

where T_u is the number of events for entity u , τ_t^u is the event time, and x_t^u contains the scope-specific public fields observable at that time. We omit the entity superscript u when it is unambiguous.

Events follow a fixed chronological order with deterministic tie-breaking. Before processing e_t^u , an agent has a causal state Z_t^u constructed only from events and finalized outputs preceding e_t^u in this order. It produces

$$(\hat{y}_t, \hat{m}_t) = \pi(e_t^u, Z_t^u), \quad \hat{y}_t \in \{0, 1\}, \quad (2)$$

where $\hat{y}_t = 1$ denotes INTERVENE, $\hat{y}_t = 0$ denotes ABSTAIN, and $\hat{m}_t$ is an optional evidence-grounded service message. Here π denotes a general inference policy over the current event and causal state; $\hat{m}_t$ may be empty under ABSTAIN. After finalizing the decision and optional message, the agent updates its state through an update function U :

$$Z_{t+1}^u = U(Z_t^u, e_t^u, \hat{y}_t, \hat{m}_t). \quad (3)$$

Each entity maintains isolated state, and π accesses only the current public event and causally available same-entity information.

The objective is to identify a timely, actionable *service opportunity* supported by the current event or prior same-entity observations. Let $y_t \in \{0, 1\}$ denote the frozen, reviewed reference decision for e_t under the annotation protocol in Section 4.2; all evaluated policies share these fixed references. Labels and annotation metadata remain evaluator-private. Events with either reference label share the same scope-specific public input schema.

3 Proactive Memory (ProactMem)

ProactMem uses three complementary memory layers (Figure 2): Event Memory for recent public events, Daily Trajectory Memory for source-checked daily abstractions, and Temporal Pattern Memory for replay-assessed recurring patterns. Construction checks source fidelity and historical support, retrieval selects currently applicable patterns, and the decision LLM assesses present service value.

3.1 Memory Construction and Update

Bounded Event Memory. For event e_t , Event Memory retains prior events in their original public representation, intersecting a temporal horizon Δ with capacity K :

$$S_t = \text{Tail}_{K-1}((e_i)_{i < t, 0 \leq \tau_t - \tau_i \leq \Delta}). \quad (4)$$

Here Tail_{K-1} selects the most recent $K - 1$ events in the filtered sequence. We use $\Delta = 72$ hours and $K = 100$: at most 99 prior events accompany the separately supplied current event. An independent daily buffer retains all committed events of the open local day, preserving complete input for daily construction.

Fact-validated Daily Trajectory Memory. After a local day closes, a construction LLM summarizes its buffered events as trajectory segments with intervals, typed facts, and event citations. Let $F(\cdot)$ denote normalized checkable facts obtained from typed fields. For segment d and its cited events $\mathcal{E}(d)$, source consistency requires

$$F(d) \subseteq \bigcup_{e \in \mathcal{E}(d)} F(e), \quad \mathcal{E}(d) \subseteq E_{\text{completed day}}^u. \quad (5)$$

Here $E_{\text{completed day}}^u$ contains the same entity’s events from that completed day. The validator checks entity, interval, citations, and canonicalized scalar values at compatible field paths. Acceptance requires source-supported typed facts and resolved citations; free text provides accompanying descriptions. Accepted daily trajectories form W_t , retaining historical context beyond Event Memory. Appendix D details field handling and buffering.

Replay-assessed Temporal Pattern Memory. Weekly consolidation groups seven completed daily records per entity in time order, with each record assigned to one cycle. A cycle may span more than seven calendar days. An LLM proposes

a trigger g , expected pattern p , time window, optional weekday restrictions, and source daily identifiers. Replay checks these conditions against validated daily facts: an observable triggered day is fulfilled when p occurs at or after the trigger within the window and violated otherwise. Missing observations are censored, inapplicable days are excluded, and each date contributes at most once. Let n_f count fulfilled (supporting) days and n_v count violated days:

$$\text{conf}_{\text{replay}} = \frac{n_f}{n_f + n_v}, \quad \text{lift} = \frac{\text{conf}_{\text{replay}}}{\text{base}(p)}. \quad (6)$$

The base rate is the frequency of p over observable days under the same window/weekday restrictions across both triggered and other observable days. Zero denominators yield zero confidence or lift, respectively.

A well-formed proposal becomes *Active* when $n_f \geq 3$, confidence ≥ 0.5 , and lift ≥ 1.0 ; otherwise it remains decision-invisible *Candidate*. Malformed proposals are rejected. Consolidation resets the online daily view, retaining records for audit. Later observations update pattern confidence and lifecycle: repeated violations can lower retrieval priority or suspend eligibility. Appendix D.3 specifies these transitions. Historical support characterizes observed recurrence; future applicability and present service value require separate assessment.

3.2 Memory Retrieval and Selection

Event Memory supplies the bounded sequence and W_t supplies all completed daily trajectories in the current cycle. Pattern retrieval additionally checks entity-local trigger context, weekdays, and temporal applicability with a 30-minute lead/grace interval. Matching considers current and previously recorded causal trigger occurrences. An occurrence is fulfilled when its expected behavior has been observed; censored occurrences have insufficient observations for assessment. Selection excludes both states. For pattern store L_t^u , lifecycle eligibility Eligible_t , trigger/time matching Match_t , and occurrence state s_t ,

$$R_t = \text{Top3} \{ \ell \in L_t^u : \text{Eligible}_t(\ell) \wedge \text{Match}_t(\ell), s_t(\ell) \notin \{\text{fulfilled}, \text{censored}\} \}. \quad (7)$$

The top-three selection ranks eligible patterns by occurrence state, lifecycle, confidence, and temporal distance (Appendix D.4). All matching uses

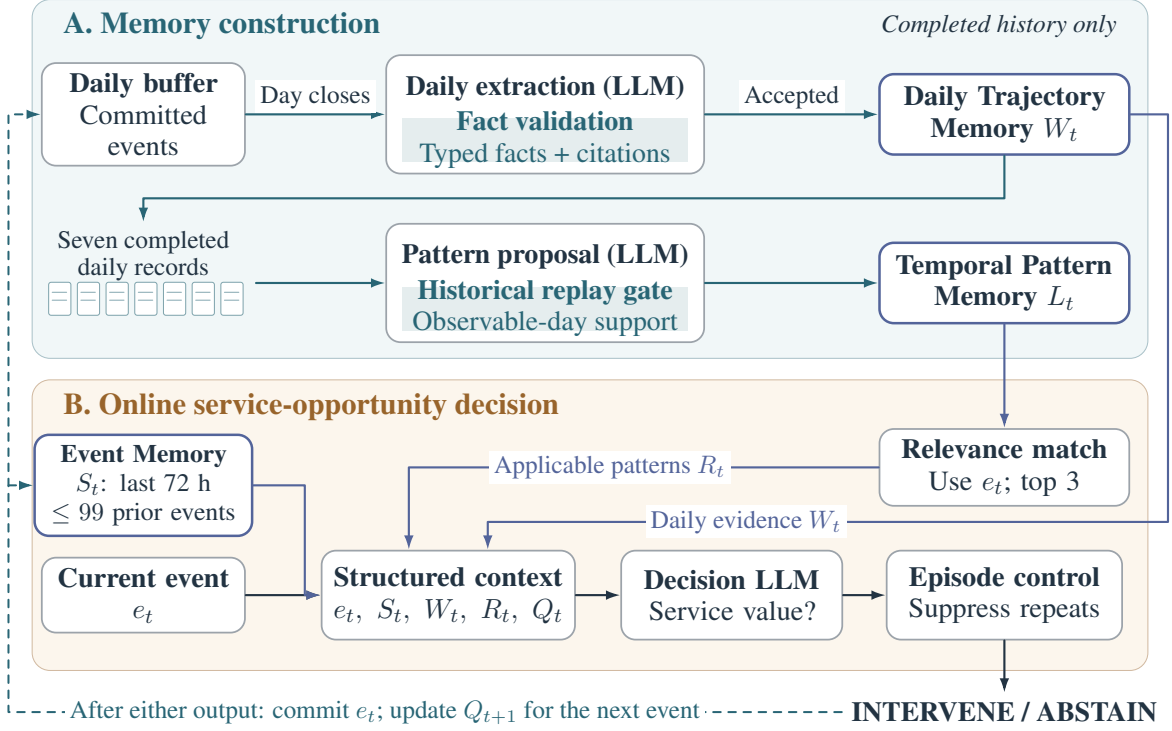


Figure 2: ProactMem separates memory construction (A) from online decision-making (B). Daily extraction is checked against cited events; temporal proposals are assessed by replay over completed daily records. Only lifecycle-eligible, currently relevant patterns enter the structured context. The decision LLM assesses service value, and the shared controller filters proposed interventions to suppress repetition. Dashed feedback commits the current event after the output is finalized, whether INTERVENE or ABSTAIN; Q_t describes observable episode state.

causally available observations to assess present applicability.

3.3 Proactive Decision

The decision context combines the current event, recent events, daily trajectories, selected patterns, and observable episode state:

$$C_t = (e_t, S_t, W_t, R_t, Q_t), \quad (8)$$

where Q_t describes the service opportunity’s observable state; the controller separately records previously allowed interventions to track service coverage. The LLM produces $(\tilde{y}_t, \tilde{m}_t) = \pi_\theta(C_t)$, proposing intervention when the current event *or* historical context supports concrete, timely, feasible, low-risk assistance, and abstention otherwise. Service value governs intervention across both routine and deviating situations.

Within each scope, all methods apply the same Episode Controller policy to their own context and intervention history. The controller compares proposed services with previously allowed interventions and converts duplicates to ABSTAIN. Scope-specific changes in opportunity state, service identity, contextual patterns, or local day permit re-

newed assessment; each intervention originates from an LLM proposal. The controller finalizes the proposal as $(\hat{y}_t, \hat{m}_t)$, the output defined in Section 2. Appendix D.5 specifies matching fields, phase transitions, and re-entry rules.

Inference prepares prior memory, stages e_t , selects relevant context, obtains and finalizes the decision, and then commits e_t . Maintenance incorporating e_t can affect only later decisions. Reference labels remain evaluator-private throughout construction and retrieval.

4 ProactiveStream Benchmark

ProactiveStream evaluates service-opportunity detection in longitudinal event streams, where agents assess incoming observations using their own accumulated history. It comprises 4,539 events across 17 isolated streams in three service scopes, with 729 reviewed reference interventions (Table 1).

4.1 Scopes and Public Events

Sports uses participant activity and availability contexts for optional walking and everyday physical-activity assistance.

Table 1: Composition and provenance of ProactiveStream across Sports, Home, and Code. Each stream is an independent memory and replay unit. Reference interventions denote team-reviewed service opportunities.

Scope	Primary source	Entity	Streams	Input events	Temporal coverage	Reference interventions
Sports	HeartSteps	participant	9	1,456	367 observed participant-days	365
Home	CASAS	home	4	1,624	14 consecutive days/home	190
Code	GH Archive	repository	4	1,459	longitudinal repository histories	174
Total	three scopes	isolated stream	17	4,539	three longitudinal settings	729

Home uses time-bounded home-sensor observations for gentle, low-risk check-ins. Each sensor segment becomes available at interval closure.

Code uses repository activity at GitHub’s native event granularity for low-risk triage, review, and maintenance, with one decision per event.

Each participant, home, or repository defines an isolated chronological stream, allowing earlier observations to inform later service opportunities. Public inputs contain contemporaneously available observations, using the same scope-specific schema for both label classes. Agents construct their own memory from these events; annotation and scoring metadata remain evaluator-private. Appendix E details source datasets, fields, and preprocessing.

4.2 Reference Labels and Causal Replay

Annotators assess e_t and its complete prior same-stream history $\mathcal{H}_{<t}$. Let A indicate a concrete, timely, feasible, low-risk opportunity supported by e_t , B an opportunity at e_t supported by historical observations, and N_t^{ref} reference-episode novelty:

$$y_t = 1 \iff (A(e_t) = 1 \vee B(e_t, \mathcal{H}_{<t}) = 1) \wedge N_t^{\text{ref}} = 1. \quad (9)$$

The reference protocol labels the first reliable actionable window INTERVENE and covered windows ABSTAIN, allowing renewed opportunities when support strengthens, the service changes, or an episode reopens. Reference novelty follows the reviewed opportunity sequence independently of model outputs and is shared across methods. The event-level labels therefore assess timely identification of reviewed service windows.

LLM-assisted screening prioritizes candidate opportunities; three team members then independently label every event while blinded to evaluated-system outputs, followed by cross-review and adjudication. Section 5.7 reports a separate-team

review of three contiguous 150-event segments with their preceding histories. Main comparisons use protocol-fixed evaluation on the jointly developed benchmark, with labels, prompts, memory bounds, and validation settings frozen before the corresponding runs.

Each stream is replayed chronologically from empty method state, with access to the current event and strictly prior same-stream context. Each decision is finalized before the current event is committed to memory; reference labels are joined after inference for scoring. Scoring includes the initial events and continues as history accumulates.

5 Experimental Evaluation

5.1 Experimental Settings

We examine overall effectiveness (**RQ1**), memory-level contributions (**RQ2**), validation mechanisms (**RQ3**), and temporal availability and decision behavior (**RQ4**).

Baselines. We evaluate five external methods:

- **ProactiveAgent** uses the released evaluation prompt and proposal/null contract with the study backbone, retaining causal event–decision history capped at 1,000 events (Lu et al., 2025; THUNLP, 2026).
- **ContextAgent-ICL** adapts ContextAgent’s in-context proactive-service judgment to public event contexts using the study backbone (Yang et al., 2025).
- **Mem0** uses open-source extraction, consolidation, and same-entity retrieval (Chhikara et al., 2025; Mem0, 2026).
- **A-MEM** constructs linked memory notes, updates their associations as events arrive, and retrieves relevant notes (Xu et al., 2025).

- **Graphiti** stores event-derived relations in a temporal graph and retrieves same-stream facts (Zep, 2026).

Summary-memory control. We separately evaluate **Cross-week Summary Memory (CSM)**, an in-house recursive-summary control, in Section 5.4. This comparison examines how retained history is represented and used for service decisions.

Memory variants. **Event-only** supplies the current event and bounded Event Memory; **Event+Daily** additionally supplies Daily Trajectory Memory. **Flat Retrieval Memory** augments Event-only with three similar older same-entity events on Sports/Home, using fixed dense embeddings and cosine similarity. All methods share the task semantics and final Episode Controller, with method-specific memory interfaces. Event-only, Event+Daily, and ProactMem also share their decision prompt, schema, and Event Memory bound. CSM, A-MEM, and Graphiti augment that same event bound with their respective representations. Appendix F records interface and retrieval settings.

Model settings. The primary model is gpt-5.6-luna, with temperature zero and thinking disabled; ProactMem uses the official OpenAI API. Terra/Sol comparisons replace both decision and memory-construction models (Section 5.6).

Metrics. We report results separately for Sports, Home, and Code using final decisions after episode control. INTERVENE includes immediate and scheduled service; ABSTAIN denotes silence at the current event. **Relevance** measures the share of interventions matching reference opportunities. **Coverage** measures the fraction of reference-positive events identified for intervention. **Effectiveness** is the harmonic mean of Relevance and Coverage. **False Positive Rate (FPR)**, also termed False Intervention Rate (FIR), is $FP/(FP + TN)$. **Balance** averages Coverage and specificity. Relevance, Coverage, Effectiveness, and Balance are task-oriented names for precision, recall, F1, and balanced accuracy, respectively; they quantify event-level agreement with the reviewed service opportunities. Appendix F provides the formulas. Higher Relevance, Coverage, Effectiveness, and Balance and lower FPR are preferred. **Intervention Rate (IR)** is the fraction of events receiving an intervention and describes decision frequency.

5.2 Main Results

ProactMem achieves the highest Effectiveness and Balance in all three scopes (Table 2, RQ1). Its Effectiveness gains over the strongest external baseline in each scope range from 4.3 to 15.7 percentage points. The comparisons below examine how opportunity recovery and intervention selectivity contribute to this service-opportunity detection performance.

Comparisons with external methods. The Effectiveness advantage spans proactive prompting, extracted memories, linked notes, and temporal-graph retrieval. On Sports, ProactMem attains the lowest FPR among all comparators; on Home, it combines the highest Relevance and Coverage. Relative to A-MEM, it recovers additional reference opportunities while reducing false interventions in every scope. These results support selecting historical information that is applicable to the current service need. The comparisons assess complete memory interfaces; the controlled ablations below examine individual design choices.

5.3 Ablation Study

Memory levels. Figure 3 compares Event-only, Event+Daily, and ProactMem with the same Luna model, Event Memory bound, decision prompt, schema, and controller. Daily trajectories improve opportunity recovery, with the largest gain on Code. Adding Temporal Pattern Memory increases Effectiveness by 9.9, 13.5, and 7.0 points on Sports, Home, and Code, respectively. The pattern layer gains true positives and reduces false positives in each scope. These complementary effects support organizing longer history into currently applicable patterns (RQ2).

The daily layer leaves false-positive counts unchanged on Sports/Home and adds two on Code, explaining its nearly flat FPR. It primarily expands recoverable context. The pattern layer adds historical support and applicability selection, improving both opportunity recovery and selectivity. Net count changes summarize aggregate differences between variants.

Validation mechanisms. Table 3 tests daily fact validation and replay gating under the same model, prompt, event bound, context capacity, and episode control. *w/o Replay Gate* admits well-formed, source-linked, online-reconstructable Candidates by bypassing Equation 6’s activation thresholds;

Table 2: Main comparison with external baselines (% , Luna). Bold marks the best directional score per scope; IR describes intervention frequency. Metrics follow Section 5.1.

Scope	Method	IR	Relevance $\uparrow$	Coverage $\uparrow$	Effectiveness $\uparrow$	FPR $\downarrow$	Balance $\uparrow$
Sports	ProactiveAgent	69.0	30.0	82.5	43.9	64.5	59.0
	ContextAgent-ICL	54.3	31.0	67.1	42.4	50.0	58.6
	Mem0	85.2	28.8	97.8	44.5	81.0	58.4
	A-MEM	60.5	30.6	74.0	43.3	56.0	59.0
	Graphiti	55.1	29.3	64.4	40.3	52.0	56.2
	ProactMem	53.8	35.8	76.7	48.8	46.1	65.3
Home	ProactiveAgent	9.2	34.2	26.8	30.1	6.8	60.0
	ContextAgent-ICL	16.1	35.9	49.5	41.6	11.7	68.9
	Mem0	8.5	23.9	17.4	20.1	7.3	55.0
	A-MEM	19.0	34.6	56.3	42.9	14.1	71.1
	Graphiti	17.9	33.4	51.1	40.4	13.5	68.8
	ProactMem	17.7	48.6	73.7	58.6	10.3	81.7
Code	ProactiveAgent	53.1	17.5	78.2	28.7	49.7	64.2
	ContextAgent-ICL	32.0	22.5	60.3	32.8	28.2	66.1
	Mem0	7.7	39.8	25.9	31.4	5.3	60.3
	A-MEM	30.4	30.4	77.6	43.7	24.0	76.8
	Graphiti	28.1	26.8	63.2	37.7	23.3	69.9
	ProactMem	26.7	40.1	89.7	55.4	18.1	85.8

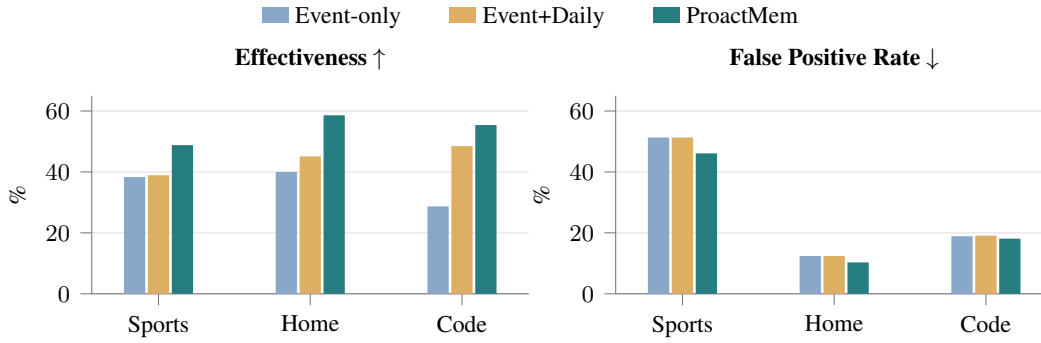


Figure 3: Memory-level ablation (% , Luna). Daily trajectories increase Effectiveness while leaving FPR nearly unchanged; temporal patterns further increase Effectiveness and reduce FPR in every scope. Model, prompt, event bound, and controller are shared. Gains are computed before rounding. Appendix G reports exact scores and net TP changes.

current-applicability matching remains enabled. *w/o Daily Validation* passes schema-valid, same-entity daily extractions directly to storage, retaining weekly construction and replay.

Both checks improve Relevance and Effectiveness across scopes (RQ3). Replay gating produces the larger Effectiveness contribution and improves Coverage in every scope. Under a fixed context capacity, replay-based admission may improve Coverage by prioritizing historically supported patterns for the decision model. On Code, ProactMem produces 43 fewer false interventions than *w/o Daily Validation* while retaining the same true-positive count. This comparison supports the contribution of source-checked daily extractions to downstream selectivity.

Episode control. The separate *Before suppression* diagnostic measures the controller’s action on recorded Code proposals. Suppression removes 66 false-positive and 17 true-positive proposals, increasing Relevance and Effectiveness while lowering event-level Coverage. This operating-point change concerns duplicate-opportunity control; the sequential memory states and later decisions remain fixed in this diagnostic.

5.4 Comparison with Recursive Summarization

We use our in-house CSM control to examine whether ordinary recursive summaries provide a comparable alternative for using longer history.

Table 3: Validation ablations and recorded controller diagnostic (% , Luna). Relevance and Effectiveness are higher-is-better. Validation variants remove their named check. Before suppression scores recorded Code proposals while preserving the original run’s later states and decisions. Bold marks the best score; Appendix G adds IR and Coverage.

Method	Sports		Home		Code	
	Relevance	Effectiveness	Relevance	Effectiveness	Relevance	Effectiveness
w/o Replay Gate	31.1	42.7	43.6	54.4	33.8	48.1
w/o Daily Validation	34.4	47.5	45.9	56.7	36.1	51.5
Before suppression	—	—	—	—	36.7	53.6
ProactMem	35.8	48.8	48.6	58.6	40.1	55.4

Construction and use. Within each isolated stream, CSM summarizes all committed events of each completed local day into dated prose. After seven completed observed days, it merges their summaries with the previous cross-week summary, carrying earlier context forward. Each decision receives the current event, bounded Event Memory, daily summaries awaiting consolidation, and the latest cross-week summary. Summary instructions retain recurring relationships, exceptions, and unresolved items. Events are committed after their decisions.

Comparison scope. Both methods share Luna, public streams, reference labels, the 72-hour/100-event Event Memory bound, task policy, output contract, and Episode Controller. CSM checks summary format and length; the LLM interprets factual support and current relevance. ProactMem additionally performs source checks, historical replay, and temporal applicability selection. Memory-specific guidance and construction budgets follow each implementation (Appendix F); this comparison assesses the complete memory strategies.

Results. Table 4 reports the summary comparison. CSM improves Effectiveness over Event-only in every scope; ProactMem achieves further gains in Effectiveness and Relevance. On Code, it retains 89.7% Coverage with 179 fewer false interventions and nine fewer true positives than CSM, yielding an 11.5-point Effectiveness gain. This is consistent with more selective use of retained history; Section 5.3 isolates the individual validation mechanisms.

5.5 Temporal Memory and Decision Behavior

What the construction checks retain. The Sports/Home audit covers nine participants and four homes. It links 423 daily trajectories (367 Sports and 56 Home) to 1,279 retained

source-supported segments, and records 54 unique temporal-pattern proposals: 42 Active and 12 Candidate. Table 5 separates unique artifacts from maintenance operations. The daily logs record 311 rejected validation attempts, while 22 of 140 weekly replay attempts remain below the activation gate at that update. A proposal can be assessed at multiple updates, so operation counts and final-store counts describe different units. Together, the audit and Section 5.3 answer complementary questions: the audit characterizes what is retained, while the ablations measure the decision-level contribution of the checks.

When longer history becomes usable. Figure 4a distinguishes construction, admission, and online use (RQ4). All 13 entities acquire a daily trajectory, 11 acquire an Active pattern, and eight have a pattern matched to a current event. The respective median elapsed times are 0.80, 6.79, and 10.46 days, each computed over entities reaching that stage. These are stage-specific summaries with different eligible-entity sets. Activation establishes historical support; current-applicability matching determines which admitted patterns enter a decision context. The difference in stage coverage makes both construction history and subsequent opportunities for use important to a longitudinal evaluation.

From historical pattern to current judgment. Figure 4b traces a recorded Sports decision from the 450-target diagnostic run, selected for its linked source events, construction artifacts, and matching telemetry. STILL observations on December 15, 17–19, and 21 become source-linked daily trajectories. The December 22 weekly update activates an afternoon pattern citing those five days, applicable during 16:00–22:00. At the matched December 24 event, the context contains 12 prior events, two daily trajectories, and that pattern; the current ob-

Table 4: Recursive-summary comparison (% , Luna), using the same full-scope inventories as Table 2. Bold marks the better directional score in each pair; IR describes intervention frequency.

Scope	Method	IR	Relevance $\uparrow$	Coverage $\uparrow$	Effectiveness $\uparrow$	FPR $\downarrow$	Balance $\uparrow$
Sports	CSM	58.9	30.8	72.3	43.2	54.4	59.0
	ProactMem	53.8	35.8	76.7	48.8	46.1	65.3
Home	CSM	17.1	43.7	63.7	51.8	10.9	76.4
	ProactMem	17.7	48.6	73.7	58.6	10.3	81.7
Code	CSM	39.5	28.6	94.8	43.9	32.1	81.4
	ProactMem	26.7	40.1	89.7	55.4	18.1	85.8

Table 5: Memory-construction audit for Sports and Home (nine participants and four homes). Log rows count maintenance operations; store rows count unique retained artifacts.

View	Counting unit	Audit result
Daily validation log	validation attempt	311 rejected before storage
Retained daily store	unique evidence-linked segment	1,279 checked; all source-supported; 0 cross-entity
Weekly replay log	replay attempt	140 attempts; 118 pass; 22 remain below the gate at that update
Retained pattern store	unique temporal-pattern proposal	54 checked; 42 Active; 12 Candidate; 0 inconsistent re-plays

servation records 256 steps in the preceding 30 minutes and 778 in the preceding hour. The model returns ABSTAIN, and the controller preserves it before the event is committed. STILL describes activity at the recorded historical observations, while the current step counts provide additional context for assessing service value. This trace illustrates the separation between pattern applicability and the final intervention judgment; Appendix B retains the source identifiers and exact timestamps.

Interpreting observable conditions. The recorded Event-only error cases show why a service decision requires context: a Sports activity suggestion follows substantial preceding movement, while a Home check-in accompanies ordinary bedroom–bathroom movement. The corresponding missed opportunities involve a reference-positive low-activity event and a quiet period spanning two Home segments. These cases illustrate the need to combine current observations, historical context, and a concrete service need. They provide qualitative context for the controlled memory-level comparisons; the source-linked case inventory remains in Appendix B.

Intervention frequency and selectivity. For reference-positive prevalence ρ , intervention frequency decomposes as

$$IR = \rho \text{ Coverage} + (1 - \rho) \text{ FPR}. \quad (10)$$

All quantities here are proportions: the first term counts recovered reference opportunities and the second counts false interventions. Home and Code have similar positive prevalence (11.7% and 11.9%), yet ProactiveAgent intervenes at 9.2% and 53.1%, respectively. Its Coverage and FPR also rise on Code, showing a more expansive operating point on repository activity. In the same scope, ProactMem combines higher Coverage with lower FPR and an IR of 26.7% (Table 2). Thus, the service setting and decision selectivity jointly shape IR; Coverage and FPR explain whether the frequency reflects opportunity recovery or excess intervention.

5.6 Retrieval and Model Sensitivity

Retrieving older events. Flat Retrieval Memory adds the three most similar older same-entity events to Event-only, using the configuration in Section 5.1. Its Effectiveness reaches 41.3% on Sports and 48.2% on Home, compared with 48.8% and 58.6% for ProactMem. Older observations improve on the bounded Event-only context, while the full hierarchy yields additional gains. Alongside the CSM comparison, this result supports examining how retained history is organized and selected for the current decision.

Changing construction and decision models. Table 6 replaces the decision LLM and the LLMs constructing daily trajectories and temporal patterns together. The benchmark, architecture, role-

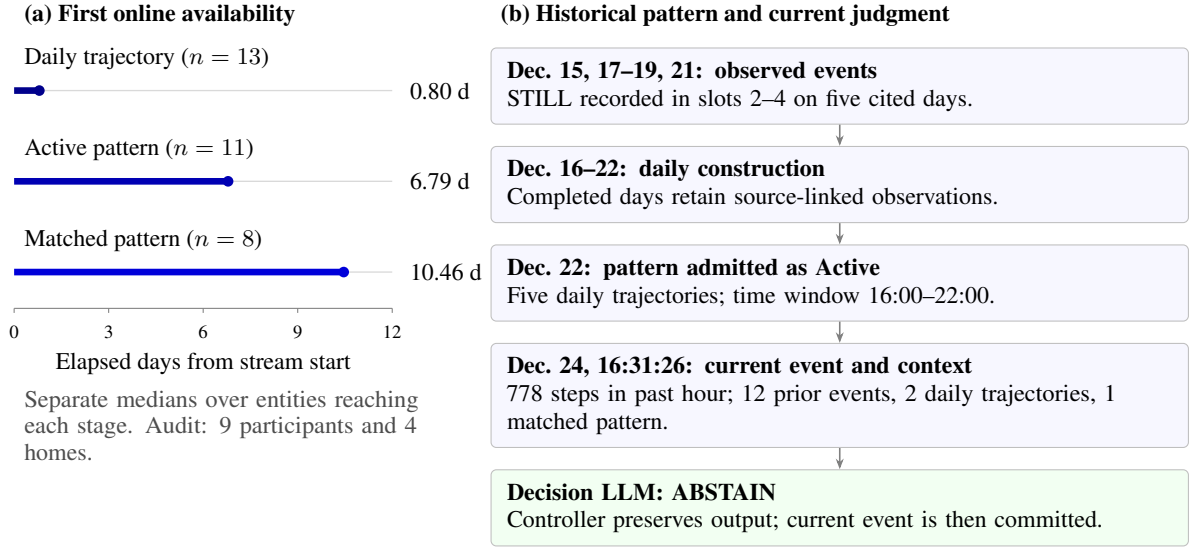


Figure 4: Memory availability and a recorded decision. (a) Sports/Home medians use the entities reaching each stage ($n = 13$, 11, and 8); each point is a separate stage summary. (b) A Sports trace from the selected-segment diagnostic run links historical observations, daily construction, an Active pattern, and a later decision. Dates are in December 2015, with stored UTC timestamps. The observed ABSTAIN originates from the decision LLM and passes through episode control unchanged.

Table 6: ProactMem model sensitivity (%). Decision and construction LLMs change together. Bold marks the best value per scope.

Scope	Model	Effectiveness $\uparrow$	FPR $\downarrow$
Sports	Luna	48.8	46.1
	Terra	54.1	39.4
	Sol	58.6	34.4
Home	Luna	58.6	10.3
	Terra	64.4	9.1
	Sol	69.3	7.5
Code	Luna	55.4	18.1
	Terra	60.3	14.9
	Sol	66.4	11.8

specific prompts, decoding settings, validation thresholds, and controller remain fixed; the resulting memory content can change. From Luna to Terra to Sol, Effectiveness increases and FPR decreases in every scope. This is a pipeline-level comparison within one model series, including model-dependent changes in both memory content and decision reasoning. All controlled hierarchy and validation ablations retain Luna.

5.7 Reviewed Labels and Episode Behavior

Sampling and reference review. We select 150 consecutive targets from one isolated stream per scope, retaining each segment’s complete preceding same-stream history. A fixed SHA256-based

rule selects eligible streams and contiguous starts from public metadata, requiring at least seven days of prior history. The 37 Sports, 203 Home, and 152 Code prefix events provide causal context for 450 scored targets. A separate three-reviewer team labels the targets with model outputs hidden and discusses its judgments to form consensus references. The review yields 27, 15, and 34 reference interventions in Sports, Home, and Code. Agreement with the earlier references is 88.7%, 91.3%, and 85.3%, respectively. Sports and Home remove 13 reference interventions each and add four and zero; Code removes six and adds 16. These label-version comparisons indicate broad agreement together with meaningful differences at service-opportunity boundaries. Appendix C gives the selected streams, periods, and confusion counts.

Holding model decisions fixed. Table 7 scores the same recorded decisions against both reference versions for each method. The outputs come from the selected-segment diagnostic runs, separate from the full-scope runs, using Luna through the official API with temperature zero and thinking disabled. The revised references leave the main benchmark labels and results unchanged. ProactMem retains its Effectiveness advantage over CSM on Home and Code under both versions; the small Sports difference changes direction. For Home, its Effectiveness changes from 68.7% to 55.6% solely through

Table 7: Effectiveness (%) under two reference versions, using identical diagnostic outputs for each method (150 targets per scope). Earlier references are original Sports/Home labels; Code combines 122 original and 28 subsequently completed human labels.

Scope	Method	Earlier Effectiveness	Reviewed Effectiveness
Sports	CSM	44.4	41.0
	ProactMem	42.6	42.5
Home	CSM	46.4	37.2
	ProactMem	68.7	55.6
Code	CSM	49.4	46.5
	ProactMem	57.6	47.4

the reference change, whereas CSM changes from 46.4% to 37.2%. This paired view makes reference sensitivity visible while holding the segment and model outputs constant.

Episode-level scoring. We additionally examine the same decisions with a common inventory of 67 proposed service episodes: 26 Sports, 10 Home, and 31 Code. An episode specifies a service goal and object, onset, valid observed positions, and an explicit deadline or closure where available. Each method initializes coverage from its own prefix interventions. The first timely matching intervention covers the episode; a later match is valuable when new causal information changes the actionable service and the message addresses that change. Continued status or rephrased advice is counted as repetition. Scheduled interventions use their intended service times when checking explicit deadlines. Each output credits at most one episode under a fixed reference order, and each episode contributes once to coverage.

Let I count all target interventions, F first valid interventions, V valuable follow-ups, and R redundant follow-ups. Episode Coverage is the fraction of new reference episodes covered at least once; Supported Relevance (SR) is $(F + V)/I$; Redundant Intervention Rate (RIR) is R/I . These denominators include unmatched, uncertain, late, and unresolved interventions in I . This diagnostic uses model-assisted episode references and semantic matches, assessed separately from the blinded event-label review. Six Home follow-up classifications have human confirmation; the remaining references and matches retain provisional status.

Coverage, follow-up, and repetition. Table 8 separates the behavioral dimensions. In Sports, both methods cover 23 of 26 proposed oppor-

tunities, with 86 interventions from ProactMem and 90 from CSM. In Home, ProactMem covers two more opportunities and provides six valuable follow-ups versus three, alongside a higher RIR. In Code, CSM covers more proposed opportunities, while ProactMem achieves higher SR with fewer interventions. The comparison therefore identifies different coverage-selectivity trade-offs under the shared inventory. The modest supported shares reflect the proportion of messages matched to that inventory; unmatched messages remain available for semantic review. These observations describe service-opportunity alignment in the sampled streams; prospective assessment is needed to establish recipient benefit. Appendix C.1 retains the outcome decomposition, boundary cases, and matching details.

6 Related Work

Agent memory and long-horizon retrieval. Generative Agents, MemoryBank, and MemGPT organize observations and experience through reflection, updating, or tiered context management (Park et al., 2023; Zhong et al., 2024; Packer et al., 2023). Mem0, A-MEM, and HippoRAG develop extracted memories, linked records, and graph-based retrieval (Chhikara et al., 2025; Xu et al., 2025; Gutiérrez et al., 2024). Graphiti maintains event-derived relations with temporal validity and provenance for retrieval (Zep, 2026). ProactMem derives recurring conditional patterns from daily trajectories, assesses their historical support, and matches applicable patterns to incoming events.

Proactive decision and memory. ProactiveAgent studies whether an agent should propose a task or remain silent and uses annotated event data to elicit this capability (Lu et al., 2025). ContextAgent combines sensory contexts with historical personas to anticipate service needs (Yang et al., 2025), while PASK couples streaming demand detection with hybrid long-term memory (Xie et al., 2026). Memory-oriented proactivity also supports internal reasoning and execution: MemCog integrates proactive memory exploration into conversational reasoning (Li et al., 2026), and Remember When It Matters uses a separate memory agent to inject selected reminders into long-horizon action agents (Wu et al., 2026). ProactMem focuses on constructing service-relevant temporal memory through source

Table 8: Paired episode diagnostic on 150 targets per scope with partially confirmed references and matches. Both methods use the same references and their own causal histories. Rates are percentages. SR credits first valid assistance and valuable follow-up; RIR counts redundant follow-up. Bold marks the better unrounded directional value within each scope (or both for ties).

Scope	Method	Covered / New	Episode Coverage $\uparrow$	SR $\uparrow$	RIR $\downarrow$
Sports	CSM	23/26	88.5	26.7	1.1
	ProactMem	23/26	88.5	27.9	1.2
Home	CSM	8/10	80.0	39.3	32.1
	ProactMem	10/10	100.0	41.0	43.6
Code	CSM	18/31	58.1	33.8	4.6
	ProactMem	14/31	45.2	40.5	9.5

checking, historical replay, and current-event matching.

Evaluation of proactive service. LoCoMo and LongMemEval assess long-term conversational memory and reasoning (Maharana et al., 2024; Wu et al., 2025). Proactive benchmarks examine task proposals, latent-need detection, or memory triggering (Lu et al., 2025; Xie et al., 2026; Li et al., 2026). ProactiveStream evaluates sequential INTERVENE/ABSTAIN decisions as agents accumulate memory within isolated, causally replayed event streams, using episode-aware references to assess intervention timing. Research on notification timing further shows that interruption cost depends on task phase and receptivity (Adamczyk and Bailey, 2004; Iqbal and Bailey, 2007, 2008; Mehrotra et al., 2016). This motivates evaluating opportunity recovery together with unnecessary intervention.

7 Conclusion

We presented ProactMem, a proactive memory framework that combines bounded recent events, source-checked daily trajectories, and replay-assessed temporal patterns. Across the three scopes of ProactiveStream, ProactMem achieves higher Effectiveness and Balance than the evaluated baselines and cross-week summary control. Ablations support the contributions of hierarchical organization, daily fact validation, and historical replay gating. These findings highlight the value of matching source-grounded, historically supported memory to current observations for service-opportunity detection.

Limitations

ProactiveStream covers three low-risk service scopes and 17 streams under a protocol fixed after joint method and benchmark development. In-

dependent held-out evaluation and cross-family model transfer remain future work. Reference labels represent reviewed opportunities; the 450-target review assesses three selected streams, while episode judgments remain partially confirmed. The Code suppression diagnostic holds later states fixed; controller-free behavior requires a sequential comparison. Historical replay assesses observed support; prospective validity under changing behavior requires further study. Our offline metrics evaluate opportunity detection. Recipient benefit, interruption burden, end-to-end cost, and deployment safety require prospective measurements. The reported results support comparisons within the evaluated protocols, with broader generalization depending on additional populations, observation conditions, and model families.

Ethical Considerations

Our evaluation is limited to optional activity support, gentle check-ins, and low-risk repository assistance. Medical and high-risk safety decisions require separate domain-specific validation and safeguards. Inference is isolated by stream, and reference labels, episode annotations, and scoring metadata are excluded from model inputs.

Longitudinal events and derived memories can reveal sensitive routines even when identifiers are pseudonymized. Event content, prediction traces, and inferred trajectories require access and retention protections; external-model use should minimize transmitted information and respect source-data permissions.

Deployment should require informed opt-in and allow inspection, correction, deletion, retention control, and disabling proactive assistance. An intervention prediction represents a service proposal; consequential action requires separate authoriza-

tion. These deployment safeguards require separate implementation and evaluation.

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A Memory-Construction Audit and Online Availability

Complementing the evidence-check ablations in Section 5.3, we audit existing memory-construction artifacts and maintenance logs from Sports and Home, covering nine participants and four homes. The main-text audit in Table 5 distinguishes validation and replay operations from unique retained artifacts; this appendix records counting and lifecycle details. The audit checks source support, entity consistency, and replay consistency.

Retained evidence. The daily store contains 423 trajectories, with 367 from Sports and 56 from Home, comprising 1,279 retained segments. All retained segments pass the configured source-support and entity-consistency checks, with all evidence linked to the same entity. Separately, construction logs record 311 rejected daily validation checks. These observations characterize construction errors, their filtering, and the resulting retained evidence.

The final temporal-pattern store contains 54 source-linked proposals whose recorded replay results are consistent with offline recomputation: 42 are Active and 12 remain Candidate. Decision contexts use lifecycle-eligible patterns that satisfy current-applicability conditions (Appendices D.3 and D.4). Across weekly maintenance updates, 118 of 140 replay attempts pass the activation gate, while 22 remain below its thresholds. These operation counts are distinct from the 42 unique Active patterns in the final store.

Online availability. Daily Trajectory Memory first becomes available for all 13 entities at a median elapsed time of 0.80 days. An Active temporal pattern first becomes available for 11 entities at a median of 6.79 days. Current-applicability matching first supplies a temporal pattern for eight entities at a median of 10.46 days. Each median is calculated over entities reaching the corresponding stage. Pattern activation and current relevance are checked separately: admitted patterns enter a decision context after current-applicability matching.

These observations show that daily and temporal-pattern evidence can become available within the evaluated streams, leaving subsequent events for evidence-informed decisions. The update order in Section 3.3 preserves the causal boundary: evidence used to judge an event is constructed from previously committed events. The construction checks complement the RQ3 validation ablations

in Section 5.3; the stage-specific availability statistics address RQ4.

B Illustrative Decisions and Error Cases

Table 9 traces one recorded Sports decision from the 450-target diagnostic run, selected because its source events, daily artifacts, contemporaneous weekly proposal, and matched-pattern telemetry can be linked. It uses the proposal version available at decision time. The example illustrates evidence flow through one recorded decision.

The current step counts provide contemporaneous activity evidence for assessing the earlier STILL pattern’s service relevance. We interpret the logged decision using the recorded current observations and matched pattern. The logged ABSTAIN originates from the decision LLM and passes through the controller unchanged, illustrating separate pattern-applicability and intervention judgments.

The separate frozen 16-case inventory includes two false positives and two false negatives from Event-only logs. In Sports, the false-positive case receives an activity suggestion following substantial movement during the preceding hour; the false-negative case receives ABSTAIN at a reference-positive low-activity event. In Home, the false-positive case receives a check-in during ordinary bedroom–bathroom movement; the false-negative case receives ABSTAIN during a reference-positive quiet period spanning two consecutive segments.

These errors illustrate the distinction between an observable condition and an actionable service opportunity. These cases motivate joint assessment of current activity, historical context, and actionable service needs. Historical evidence contextualizes these observations for the agent’s final service-value judgment. Repeated windows additionally require assessing whether evidence or service needs have changed, complementing the controller trade-off examined in Section 5.3.

This qualitative analysis addresses the failure-analysis part of RQ4 and illustrates why Proact-Mem separates memory evidence, present service value, and prior opportunity coverage.

C Independent Human Review Across Three Scopes

We conduct a sampled label review within the main benchmark, selecting 150 consecutive target events per scope (450 in total). A separate three-reviewer

Table 9: A source-to-decision timeline for one Sports participant. Dates are in December 2015; times use the stored UTC timestamps. Text condenses source events and construction/decision logs.

Stage and recorded time	Source-linked observation or operation
Historical events Dec. 15, 17–19, 21	Participant 35 has recorded STILL observations in slots 2–4 across five cited days. For example, Dec. 15 at 16:32:38 records slot 2, work, STILL, and 586 steps in the preceding hour (event 5fd72a80. . .). STILL describes activity at the recorded observation time.
Daily construction Dec. 16–22	Each cited day becomes a source-linked daily trajectory at the next daily rollover. The Dec. 15 trajectory retains the example’s event reference and observed fields; its recorded creation time is Dec. 16 at 12:02:49.
Weekly proposal Dec. 22	At the weekly update on Dec. 22, the proposal cites the five completed trajectories dated Dec. 15, 17–19, and 21: recognized activity repeatedly equals STILL in slots 2–4, with applicability window 16:00–22:00. The maintenance record at 12:02:27 reports one proposed and one activated pattern.
Current event Dec. 24, 16:31:26	Event c956aa51. . . : slot 2, available, in-transit status false, store location, activity UNKNOWN, 256 steps in the preceding 30 minutes and 778 in the preceding hour. The context contains 12 prior events, two daily trajectories, and the matched pattern 93463c34. . . .
Decision, then commit	Both model proposal and final output are ABSTAIN; controller action is ALLOW, preserving the model output. Telemetry records that the current event is committed after context construction. Here, the matched recurring observation accompanies an ABSTAIN decision.

team independently annotates these segments while blinded to ProactMem and CSM outputs, and discusses its judgments to obtain consensus references. Each segment belongs to exactly one participant, home, or project repository. A fixed SHA256-based rule first ranks eligible streams and then eligible contiguous starts using public metadata only, and requires at least seven days of preceding history. The selected streams are Sports participant_35, Home home_0004, and the Code repository callstack/react-native-pager-view. Their target periods are December 16, 2015–January 18, 2016; June 23–28, 2011; and June 21–November 4, 2025, respectively. The corresponding ProactMem pipeline processes all 37, 203, and 152 preceding events chronologically to provide historical context; scoring covers the 150 target events in each scope. Thus, 842 public events supply causal context for 450 scored targets; each target can use only its current observation and strictly prior same-stream history.

The reviewers inspect each target together with its preceding same-stream history. The confirmed review yields 27/123, 15/135, and 34/116 INTERVENE/ABSTAIN labels for Sports, Home, and Code, respectively. Agreement with the earlier diagnostic references is 88.7% (133/150), 91.3% (137/150), and 85.3% (128/150) for Sports, Home, and Code, respectively. In Sports and Home, 13 and 13 labels change from INTERVENE to ABSTAIN, compared with 4 and 0 changes in the reverse direction; Code has 6 removals and 16 additions. Most judgments agree, with consequential differences at service-

opportunity boundaries. The reported counts quantify agreement between label versions and the directions of label changes. These annotations serve as references for the additional evaluation; the main benchmark labels and result tables remain unchanged. For model comparison, we use the recorded ProactMem and CSM decisions from the selected-segment diagnostic runs, which are separate from the main runs. For each method, exactly the same target decisions are scored against both label versions; rescoring preserves the recorded outputs. Like the main experiment, this diagnostic uses gpt-5.6-luna through the official OpenAI API, with temperature zero and thinking disabled. Decision-output limits are 256 tokens for Sports/Home and 512 for Code. Both methods retain the 72-hour/100-event bound and episode control; ProactMem additionally retains daily validation and replay gating.

Section 5.7 reports the paired label-version comparison in Table 7; Table 10 below retains ProactMem’s detailed counts under the confirmed event-level references. ProactMem identifies 24/27 Sports, 15/15 Home, and 18/34 Code opportunities. Its Effectiveness remains above CSM in Home and Code under both versions; the small Sports difference changes direction. Thus, the paired audit reveals both agreement and label-sensitive method rankings. For example, Home ProactMem Effectiveness changes from 68.7% to 55.6% when only the references change, while CSM changes from 46.4% to 37.2%. These comparisons hold the selected segment and its diagnostic outputs fixed; the

main evaluation uses its own full-scope events and runs. Across scopes, immediate and scheduled service decisions share the INTERVENE label. The paired episode analysis in Appendix C.1 uses references grounded in these reviewed events and their causal histories, with its annotation status specified below.

Table 10: Recorded ProactMem decisions evaluated against consensus labels from a team blinded to model outputs, on 450 targets with 150 contiguous decisions per scope. Each segment and its prefix are entity-isolated. Counts use final intervention decisions after episode control, including immediate and scheduled outputs. Prefix events supply historical context.

Quantity	Sports	Home	Code
Scored decisions	150	150	150
Context-only preceding events	37	203	152
Reference interventions	27	15	34
Reference abstentions	123	135	116
ProactMem interventions	86	39	42
TP	24	15	18
FP	62	24	24
FN	3	0	16
TN	61	111	92
Intervention Rate	57.3%	26.0%	28.0%
Relevance	27.9%	38.5%	42.9%
Coverage	88.9%	100.0%	52.9%
Effectiveness	42.5%	55.6%	47.4%
False Positive Rate	50.4%	17.8%	20.7%
Balance	69.2%	91.1%	66.1%

C.1 Paired Episode-Level Behavioral Diagnosis

On the same 450 targets and 392 preceding events, we compare recorded CSM and ProactMem decisions using a common set of 67 proposed episodes: 26 in Sports, 10 in Home, and 31 in Code. The analysis separates first effective assistance, useful follow-up, and repetition, using fixed recorded outputs and event labels. Each episode specifies a service goal and object, causal evidence, onset, valid observed event positions, and any explicit deadline or closure evidence. This post-hoc paired diagnostic uses model-assisted episode references and semantic matches, assessed separately from the blinded event-label review. Six Home follow-up classifications have human confirmation; the remaining episode and matching proposals retain provisional status. Validity is assessed at the listed observations in each episode reference.

Reference and model histories. Fixed event labels use reference-opportunity novelty (Equa-

tion 9); episode coverage follows the evaluated method’s actual preceding interventions. Each method initializes coverage from its own prefix outputs. Scoring counts only target interventions, and each method maintains an independent coverage history. A target intervention must match an episode’s service goal and object at a listed valid observation. For timeliness, intended service time is the decision time plus the specified delay, with zero delay for immediate intervention. If an explicit reference deadline exists, coverage credit requires the intended service time to meet that deadline. The first timely match covers the episode. For subsequent matches, we compare the message and available evidence with that method’s preceding assistance for the same episode. A follow-up is valuable when new causal evidence changes the actionable service or next step and the message addresses that change; rephrasing the same advice or continued status alone is redundant. Cases requiring unobserved delivery, recipient response, or task completion are left uncertain. Valuable follow-ups contribute to service quality within the existing episode; coverage counts each episode once. Each output credits at most one episode: among compatible references, the first uncovered one in the fixed reference order takes precedence; otherwise the first already covered one is used.

Metrics. Let I be all target interventions, F first valid interventions, V valuable follow-ups, and R redundant follow-ups. Episode Coverage is covered new episodes divided by all new episodes; its unit is the service episode, whereas main-table Coverage uses reference-positive events. Supported Relevance (SR) is $(F + V)/I$, extending the first-only Episode Relevance F/I while retaining single-credit episode coverage. Redundant Intervention Rate (RIR) is R/I . All target interventions contribute to I in SR and RIR, including uncertain follow-ups (U), unmatched outputs, late outputs, and unresolved matches. Their numerators count supported assistance ($F + V$) and redundant follow-ups (R), respectively; Episode Coverage uses the number of new reference episodes as its denominator. These categories describe compatibility with the proposed references. Recipient benefit requires prospective validation. The main-text Table 8 reports coverage and rates; Table 11 below retains the count decomposition.

Observation boundaries. The main diagnostic treats the first Home opportunity as target-onset; a

Table 11: Intervention counts underlying Table 8. I equals the sum of all seven outcome categories. U denotes uncertain follow-up value; Unresolved denotes a pending semantic or condition match. Both remain separate from redundant or supported assistance. Each episode contributes once to the episode count.

Scope	Method	I	F	V	R	U	Unmatched	Late	Unresolved
Sports	CSM	90	23	1	1	0	64	0	1
	ProactMem	86	23	1	1	0	60	0	1
Home	CSM	28	8	3	9	0	8	0	0
	ProactMem	39	10	6	17	0	6	0	0
Code	CSM	65	18	4	3	3	30	6	1
	ProactMem	42	14	3	4	0	16	4	1

plausible continuation from the prefix changes both methods’ denominator to nine, with CSM covering 7/9 and ProactMem 9/9. Coverage describes the observed segment; lifetime coverage requires follow-up beyond that segment. Code has six late CSM outputs and four late ProactMem outputs; scheduled decisions are assessed using their intended service times and explicit deadlines. Three CSM Code follow-ups have uncertain value: accepting all three would raise its SR from 33.8% to 38.5%, a sensitivity scenario for those three judgments. Alternative service interpretations of other unmatched and unresolved outputs remain open for review.

Behavioral comparison. In Sports, both methods achieve the same coverage, with fewer interventions from ProactMem. In Home, ProactMem covers two more proposed opportunities and provides more valuable follow-ups. Its redundant-intervention rate is also higher. In Code, CSM covers more opportunities. ProactMem achieves higher SR with fewer interventions. The supported shares measure alignment with this finite reference inventory; the value of remaining messages requires additional semantic review. These trade-offs characterize coverage, supported follow-up, and repetition under the shared diagnostic rubric. Controller-specific causal effects require a controlled sequential comparison.

D Memory Implementation Details

This appendix expands the evidence lifecycle in Section 3; the core acceptance conditions and causal interfaces remain in the main paper.

D.1 Daily Buffer and Checkable Fields

Daily construction uses an independent buffer that preserves all committed events of the open local day alongside the bounded Event Memory. The buffer is cleared after that day is processed.

Each daily segment records a local interval, typed observed, context-before, and outcome fields, and supporting event identifiers. These activity segments are distinct from service episodes used for intervention control. Scalar values are canonicalized; field paths must agree exactly or through a schema-preserving suffix. The validator also checks entity, interval, and citation consistency. Accepted facts have matching source values and resolved same-entity citations. Validation and replay operate on typed evidence; free text supplies accompanying descriptions. Accepted segments form daily trajectories retained in the current construction cycle, even after their source events leave Event Memory.

D.2 Construction Clock and Replay Boundaries

A weekly cycle is a time-ordered block of seven completed daily-trajectory records for one entity; each record belongs to one block. Cycles advance with available daily records, so a cycle may span more than seven calendar days. Empty or unobservable records advance cycle completion while leaving fulfilled and violated replay counts unchanged. After consolidation, the online daily-memory view starts a new cycle, while daily records remain available for audit.

A proposal specifies trigger context g , expected pattern p , a local-time window, optional weekday restrictions, and supporting daily identifiers. It describes a conditional behavioral expectation. Replay evaluates days satisfying the applicability conditions and assigns censored status to missing observations. An observable day with a matching trigger is fulfilled if p occurs at or after the trigger within the specified window; otherwise it is violated. Each date contributes at most once. The confidence denominator in Equation 6 includes only fulfilled and violated days. Confidence is zero when the eligible-day count is zero; lift is

zero when the base rate is zero. The base rate uses observable days under the same window and weekday restrictions, including both triggered and other observable days.

D.3 Pattern Lifecycle

Lifecycle status controls retrieval eligibility, separately from current occurrence and service-episode states. After admission, fulfilled and violated observations update stored confidence through a smoothed support ratio combining replay counts and subsequent outcomes with one initial success and one initial failure. Censored observations leave success and failure counts unchanged. Two consecutive violations move an Active pattern to *weakening*, which remains retrievable at lower priority; four move it to *dormant*. A fulfilled observation restores a weakening pattern to Active when updated confidence is at least 0.5. A dormant pattern returns only to Candidate and requires replay readmission. The retrieval-eligible lifecycle states are Active and weakening.

D.4 Current Occurrence Matching

A deterministic matcher checks trigger context, weekday restrictions, and expected windows using a 30-minute lead and grace interval. Triggers may be current or previously recorded causal occurrences. Each occurrence has one of five states: upcoming, pending, fulfilled, deviated, or censored. Retrieval considers upcoming, pending, and deviated occurrences of lifecycle-eligible patterns. These occurrences are ranked by state (deviated, pending, then upcoming), lifecycle priority, stored confidence, and temporal distance; at most three patterns enter R_t . All matching and state assessment use only the current event and causally available observations. Daily Trajectory Memory supplies all completed daily trajectories in the current cycle.

D.5 Episode Identity and Causal Execution

In Sports/Home, the controller forms an intervention key from scope-specific episode state, proposed service action and target, and sorted pattern identifiers present in the decision context. The identifier component contains the explicit pattern identifiers available in context and defaults to an empty list for plain summaries or graph facts. State includes availability/activity context for Sports and activity/room context for Home. Home additionally distinguishes onset, confirmed, and sustained

phases at one, two, and three or more consecutive matching observations, respectively. When an explicit episode key is unavailable, scope-specific time or target fields identify the opportunity, and unchanged evidence is additionally required for duplicate suppression. Changed episode state, a restarted state sequence, changed service or pattern identifiers, or a new local day can produce a different intervention key; this permits the decision model to reassess service value.

In Code, matching uses repository-object identity and the most recent allowed intervention’s mode, recipient, content, and firing condition. An exact match suppresses a repeated create proposal by converting it to ABSTAIN. Changed signatures or other episode operations pass this duplicate check. Re-entry follows these identity and phase rules, with each intervention originating from an LLM proposal. All main-table methods and memory-level variants share this policy; each method’s available evidence determines its proposal’s identity.

The causal execution order is:

1. Prepare prior memory; any due maintenance uses only previously committed events.
2. Stage e_t as the current input, defer its memory commit until the decision is finalized, and select daily and temporal evidence.
3. Obtain the decision LLM’s proposal from the structured context.
4. Apply repeated-opportunity suppression and finalize the output.
5. Commit e_t and update coverage state; maintenance incorporating e_t affects only later decisions.

Reference labels are joined only for offline scoring.

E Public Events and Reference-Label Protocol

E.1 Scopes and Public Events

Sports. This scope derives from HeartSteps mobile activity contexts (Klasnja et al., 2019). We retain 1,456 meta-events from nine complete participant streams, covering 367 observed participant-days. Complete participant streams preserve the chronology required for daily and longer-term memory construction. Public fields include

decision-time availability, recognized activity, recent step summaries, location category, weather, and local decision slot. Randomized treatment assignments and future behavioral outcomes are excluded. Reference labels concern opportunities for optional activity support.

Home. This scope derives from CASAS ambient-sensor traces (Cook et al., 2013) and contains four homes with 14 consecutive days each. Within every local day, ten-minute preprocessing blocks are merged into 28 regular 50-minute segments and one 40-minute tail, producing 406 events per home. Every source block is assigned to exactly one segment, preserving the complete two-week chronology. A segment becomes available only after its interval closes and contains within-interval motion counts, room sequences, door actions, and compact numeric summaries. Each public segment describes its own interval; the evaluated agent derives historical evidence from preceding same-home observations. The service space is restricted to gentle, low-risk check-ins and excludes disease, fall, travel, danger, and unsupported preference inference.

Code. This scope organizes 1,459 repository activity records into four isolated streams, with 174 reference interventions (Table 1). Each event represents a timestamped repository activity, such as a commit push, issue update, pull-request action, or review. We preserve GitHub’s native event granularity, with each event corresponding to one intervention decision. Within each repository, records are ordered by event time, with event ID used to break ties deterministically. The source inventory includes public GH Archive records (Grigori, 2026). Each repository independently maintains Event Memory, Daily Trajectory Memory, and Temporal Pattern Memory, together with its intervention history and episode state. Observations, extracted evidence, and coverage state remain local to each repository, including projects with shared contributors. The repository stream is the memory owner and evaluation unit. Services include low-risk triage, review, and maintenance assistance. Native PushEvent denotes repository activity; INTERVENE denotes an agent’s service decision.

Public-input contract. Reference-positive and reference-negative events follow the same scope-specific public schema. Public inputs contain only event identifiers, stream identifiers, timestamps, and contemporaneously observable fields. Refer-

ence labels, candidate flags, episode annotations, memory-level markers, and scoring masks remain evaluator-private; agents construct trajectory summaries and memory representations from public observations. The inventory includes early events observed before Daily Trajectory Memory or Temporal Pattern Memory becomes available.

E.2 Reference Labels and Causal Replay

Service-opportunity semantics. Annotators assess the current event e_t and the complete strictly prior public history $\mathcal{H}_{<t} = (e_1, \dots, e_{t-1})$ of the same entity, independently of the memory representation retained by a tested method. Let $A(e_t)$ indicate that the current event itself establishes a concrete, timely, feasible, and low-risk service opportunity, and let $B(e_t, \mathcal{H}_{<t})$ indicate that prior evidence makes such an opportunity visible at the current event. Let N_t^{ref} indicate opportunity novelty under the reference episode protocol below. The reviewed label follows Equation 9. These predicates specify annotation semantics. Annotators assess concrete service value in both routine and unusual situations, using current observations and prior evidence. Within a semantic episode, the first reliable and actionable window is labeled INTERVENE. Subsequent covered windows are labeled ABSTAIN unless the evidence strengthens, the feasible service changes, or the episode reopens. Reference episode coverage, and hence N_t^{ref} , is determined from the reviewed opportunity sequence up to the current event, independently of a method’s predicted interventions. The labels represent reviewed service opportunities.

Annotation protocol. An LLM-assisted screening stage identifies potential service opportunities and expands the review queue. Three team members then independently assign initial labels while blinded to the evaluated systems’ decisions, followed by team cross-review and adjudication. Every public event enters final review, covering the full inventory beyond the screening shortlist. The screening output prioritizes inspection; team adjudication determines each reference label. We use protocol-fixed evaluation on the jointly developed benchmark; independent held-out testing remains future work. The Home aggregation rule is specified before its event rows are constructed, and each evaluation version jointly records its event inventory, reference-label map, intervention prompt, Event Memory boundary, and validation settings.

For the main comparisons, these labels, prompts, memory bounds, and validation settings are frozen before the corresponding runs.

Causal replay. Every participant, home, or repository stream is replayed independently from empty method state and in timestamp order. At event t , a method may access the current event e_t , its own strictly prior same-stream memory, and any preceding outputs retained by its architecture. Access is restricted to the current public event and causally available same-stream state. The intervention decision is finalized before e_t is committed to memory. Daily and weekly maintenance involving e_t can therefore affect only later decisions. Reference labels remain evaluator-private during inference and are joined afterward for scoring. Annotation artifacts, reference episode states, and scoring metadata remain organizer-private.

F Additional Experimental Configuration

External baselines. *ProactiveAgent* adapts ProactiveAgent’s released evaluation prompt and proposal/null output contract (Lu et al., 2025; THUNLP, 2026), using the study backbone. It retains causal event–decision history, capped at 1,000 events including the current event per request, as its memory interface. *Mem0* uses the open-source mem0ai==2.0.12 extraction, consolidation, and retrieval pipeline (Chhikara et al., 2025; Mem0, 2026), local all-MiniLM-L6-v2 embeddings, top-8 same-entity retrieval, and a 0.1 search threshold. The baselines retain their method-specific memory and prompt interfaces while sharing the task semantics and final Episode Controller. Each method retains its configured context and token budgets.

A-MEM uses linked memory notes and association updates (Xu et al., 2025); retrieved notes augment the bounded event context. *ContextAgent-ICL* uses the in-context proactive-service formulation of ContextAgent (Yang et al., 2025), adapted to the public event representation and study backbone. These event-stream adaptations are evaluated on the same scope-specific reference labels as the other main-table methods.

Cross-week summary control. CSM is the in-house recursive-summary control evaluated in Section 5.4. It summarizes each completed local day and, after seven completed observed daily records, combines that block with the previous cross-week

summary. The block follows observed dates and may span more than seven calendar days. This recursive update combines retained information from earlier weeks with the latest block. The decision context contains the current event, bounded Event Memory, completed daily summaries in the current block, and the latest cross-week summary. The adapter caps daily and cross-week summaries at 2,000 and 6,000 characters, respectively. CSM checks output format and length before supplying the summary text directly to the decision model. Its construction inputs contain only committed public observations; prior decisions are recorded separately for controller state and resumption. Daily and cross-week versions retain source links for auditing, while the decision model interprets the summary prose for current service value.

Temporal-graph baseline. Graphiti uses the open-source temporal-graph framework (Zep, 2026) to ingest committed public events into a stream-isolated graph. The adapter uses Graphiti 0.30.2, Neo4j, and local 384-dimensional all-MiniLM-L6-v2 embeddings. The current public event is the retrieval query; native hybrid edge search with reciprocal-rank fusion returns at most ten relation facts, including validity fields and source-record identifiers. The adapter supplies the retrieved relation facts and their source identifiers directly. Graphiti’s ingestion episodes denote source records; evaluation episodes denote service opportunities. CSM and Graphiti retain Proact-Mem’s 72-hour/100-event bound for the event context and reuse the task policy, output contract, and Episode Controller with memory-specific guidance. CSM supplies summary text and Graphiti supplies relation facts through their respective memory interfaces. The comparison evaluates alternative memory interfaces with method-specific prompt guidance and computational budgets.

Memory variants and retrieval control. *Event-only* receives the current event and bounded Event Memory (Equation 4). *Event+Daily* additionally receives fact-validated daily trajectories; Proact-Mem also receives eligible temporal patterns (Equation 7). These three variants share the intervention prompt, output schema, and Episode Controller, differing in available memory evidence. For Sports and Home, *Flat Retrieval Memory* adds to Event-only the three most similar older same-entity events outside its window. Its retrieval path consists of fixed dense embeddings, cosine-similarity ranking,

and direct inclusion of the selected source events.

Mechanism ablations and controller diagnostic.

Section 5.3 defines two evidence-check ablations and a suppression diagnostic; Table 13 reports their intervention rates, Relevance, Coverage, and Effectiveness. Each validation ablation removes only its named check. The controller diagnostic compares recorded proposals with finalized outputs on the same Code sequence, holding later states and decisions fixed.

Model roles and implementation. In the primary setting, decision and memory-construction calls use gpt-5.6-luna through the official OpenAI API, with temperature zero and thinking disabled. Decision generation and memory construction have separate call roles and output budgets. Decision limits are interface-specific (256 or 512 tokens); construction-call limits are configured separately and recorded by role in the run configuration. Event Memory contains at most 99 prior events within 72 hours, plus the separately supplied current event. Daily construction follows local-day closure; weekly construction and replay follow seven completed daily records as defined in Section 3. All memory and validation records are entity-scoped.

We also test gpt-5.6-terra and gpt-5.6-sol. Both decision and construction LLMs change together; Section 5.6 provides the fixed settings and complete results. Input manifests, prompts, schemas, thresholds, and role-specific configurations are versioned in the experiment ledger.

Scoring. Section 5.1 defines the scoring policy and metrics. Using event-level confusion counts, Relevance is $TP/(TP + FP)$, Coverage is $TP/(TP + FN)$, Effectiveness is $2TP/(2TP + FP + FN)$, and Balance is $\frac{1}{2}[TP/(TP + FN) + TN/(TN + FP)]$. Effectiveness is the harmonic mean of Relevance and Coverage; the reported percentages use the same reference labels and finalized decisions as the other event-level metrics. Intervention Rate measures decision frequency.

F.1 Intervention-Rate Interpretation

Intervention Rate describes a behavioral operating point interpreted alongside coverage and selectivity. Writing ρ for reference-positive prevalence, the exact identity $IR = \rho \text{ Coverage} + (1 - \rho) \text{ FPR}$ separates opportunity recovery from false intervention, with all rates expressed as fractions. The preva-

lences are 25.1%, 11.7%, and 11.9% in Sports, Home, and Code. Scope-specific decision behavior also contributes to the variation: ProactiveAgent’s intervention rates are 9.2% on Home and 53.1% on Code, whose reference-positive prevalences are similar. Their respective Coverage and false-intervention rates reveal different coverage–selectivity operating points under the scope-specific evidence and service semantics. This descriptive decomposition summarizes the joint behavior of the LLM, memory, prompt, and controller.

G Extended Ablation and Controller Results

Tables 12 and 13 expand Figure 3 and Table 3 with net TP increments, intervention rates, and Coverage. Net TP changes summarize differences between aggregate method counts. Intervention rates and Coverage are retained to expose coverage–selectivity trade-offs, including the Code-only Episode Controller comparison.

Table 12: Memory-hierarchy ablation. Effectiveness and False Positive Rate (FPR) are percentages; Δ TP is a net count. Daily Δ TP compares Event+Daily with Event-only; Pattern Δ TP compares ProactMem with Event+Daily. The increments summarize aggregate count differences. Bold marks higher Effectiveness and lower FPR. Count increments are reported per scope.

Scope	Method	Effectiveness $\uparrow$	FPR $\downarrow$	Daily Δ TP	Pattern Δ TP
Sports	Event-only	38.3	51.3	–	–
	Event+Daily	38.9	51.3	+4	–
	ProactMem	48.8	46.1	–	+57
Home	Event-only	40.0	12.4	–	–
	Event+Daily	45.1	12.4	+15	–
	ProactMem	58.6	10.3	–	+33
Code	Event-only	28.7	18.9	–	–
	Event+Daily	48.5	19.1	+64	–
	ProactMem	55.4	18.1	–	+22

Table 13: Evidence-check ablations and controller diagnostic (%). Validation rows remove the named check; Before suppression scores proposals from the recorded Code run while holding later states fixed. ProactMem is last. Bold marks the best directional score. The controller diagnostic is reported for Code alone.

Scope	Method	Intervention Rate	Relevance $\uparrow$	Coverage $\uparrow$	Effectiveness $\uparrow$
Sports	w/o Replay Gate	55.0	31.1	68.2	42.7
	w/o Daily Validation	55.8	34.4	76.7	47.5
	ProactMem	53.8	35.8	76.7	48.8
Home	w/o Replay Gate	19.3	43.6	72.1	54.4
	w/o Daily Validation	18.9	45.9	74.2	56.7
	ProactMem	17.7	48.6	73.7	58.6
Code	w/o Replay Gate	29.4	33.8	83.3	48.1
	w/o Daily Validation	29.6	36.1	89.7	51.5
	Before suppression	32.4	36.7	99.4	53.6
	ProactMem	26.7	40.1	89.7	55.4